

Experiment 30

Write C programs to demonstrate the following thread related concepts: (i) create (ii) join (iii) equal (iv) exit.

C Program

```
#include <stdio.h>
#include <pthread.h>

void *thread_function(void *arg) {
    pthread_t self = pthread_self();

    printf("Thread created successfully.\n");
    printf("Thread ID: %lu\n", (unsigned long)self);

    printf("pthread_equal(self, self) = %d\n",
        pthread_equal(self, self));

    printf("Thread is exiting now.\n");
    pthread_exit(NULL);
}

int main() {
    pthread_t thread;

    if (pthread_create(&thread, NULL, thread_function, NULL) != 0) {
        perror("pthread_create");
        return 1;
    }

    printf("Main thread created the worker thread.\n");

    pthread_join(thread, NULL);
    printf("pthread_join completed successfully.\n");

    return 0;
}
```

Sample Output

```
Main thread created the worker thread.
Thread created successfully.
Thread ID: 140735...
pthread_equal(self, self) = 1
Thread is exiting now.
pthread_join completed successfully.
```

Result

Thread creation, joining, equality checking and thread exit were successfully demonstrated using POSIX threads.